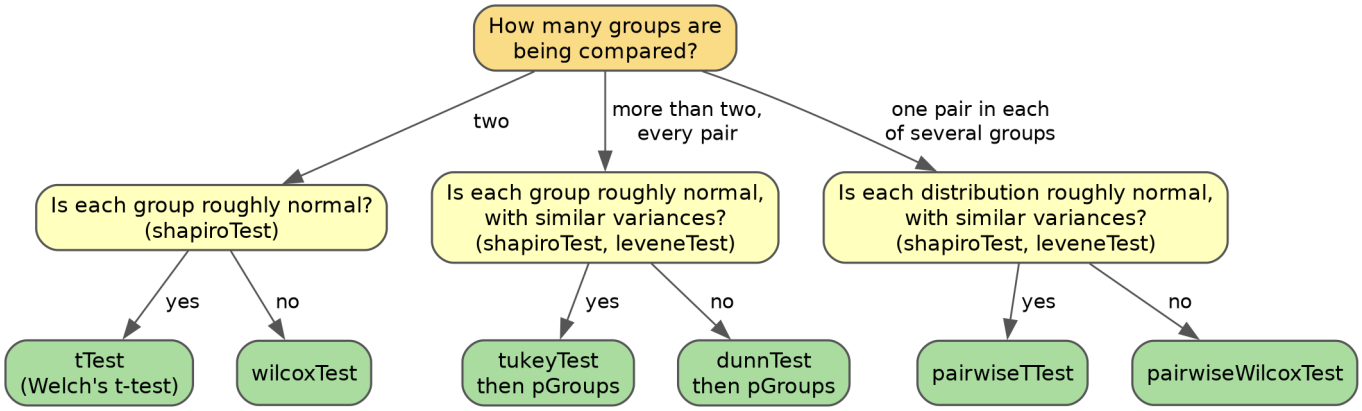
comparing means: summary

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Look at the distributions before comparing means: a mean on its own hides the shape, the spread and the number of observations. The *p* value is the likelihood of getting data this extreme if the distributions are the same, and it is not the probability that a detected difference is a false positive. Choosing a test comes down to how many groups are compared and whether the data are normal. **How many groups are being compared?** With more than two, a separate t-test on every pair multiplies the chance of a false positive, so the Tukey and Dunn tests compare every pair at once and correct the p-values (read `p.adj`). **Can each group be modeled by a normal distribution?** Check with the Shapiro test within each group. If so, use a parametric test. If not, use a rank test, which asks whether values in one group tend to be higher than values in the other. The t-test here is Welch's, so two groups need no variance test, but the Tukey test shares one estimate of the spread across groups, so check it with the Levene test, as the chapter also does before `pairwiseTTest`. A t-test and a rank test can disagree because they answer different questions, and a plot of the raw points shows which question matters.



Function	What it is for	How it is used in this chapter
`shapiroTest(abundance)`	is each group roughly normal? (p > 0.05 means not significantly different from normal)	after `group_by()` on the grouping column(s)
`leveneTest(abundance ~ aquifer_code)`	do the groups have similar variances?	before `tukeyTest` or `pairwiseTTest`
`tTest(abundance ~ aquifer_code)`	two groups, roughly normal (Welch's t-test)	`y ~ x` formula
`wilcoxTest(abundance ~ aquifer_code)`	two groups, not normal (rank test)	`y ~ x` formula
`tukeyTest(abundance ~ aquifer_code)`	more than two groups, every pair, normal with similar variances	read `p.adj`
`dunnTest(abundance ~ aquifer_code)`	more than two groups, every pair, rank-based	read `p.adj`
`pGroups()`	letters for each group from Tukey or Dunn output (a shared letter means no significant difference)	piped after `tukeyTest` or `dunnTest`, then drawn with `geom_text()`
`pairwiseTTest(abundance ~ analyte)`, `pairwiseWilcoxTest(abundance ~ analyte)`	one comparison within each of several groups	after `group_by(aquifer_code)`, with `p.adj` corrected across all the groups
`geomSignif()`	draws brackets and stars from pairwise test output	`data =` a table built from the test output